

hyper2kvm Migration Demo

Windows 11 VMware VMDK to KVM — End-to-End

Live demonstration: split VMDK flatten, SATA bootstrap boot strategy, offline registry fixes (storahci boot-critical enablement, VMware service disablement), VirtIO driver staging, libvirt domain XML generation with Hyper-V enlightenments, and successful Windows 11 OOB on KVM.

Windows 11 Home | Split VMDK 13.2 GB | QCOW2 | SATA Bootstrap | Q35/UEFI | Hyper-V | OOB Verified

Step 1-2: Source & Configuration

STEP 1 — Source VMDK (Split, 4 Extents)

```
$ qemu-img info ~/vmware/win11/win11.vmdk
image: win11.vmdk
file format: vmdk
virtual size: 15 GiB (16106127360 bytes)
disk size: 13.2 GiB
Format specific information:
  create type: twoGbMaxExtentSparse
  extents:
    [0]: win11-s001.vmdk (4.0 GiB)
    [1]: win11-s002.vmdk (4.0 GiB)
    [2]: win11-s003.vmdk (3.9 GiB)
    [3]: win11-s004.vmdk (1.4 GiB)

VMX: firmware=efi  guestOS=windows11-64  memsize=4096  numvcpus=2
```

STEP 2 — Migration YAML Config

```
cmd: local
vmdk: ~/vmware/win11/win11.vmdk
output_dir: ~/out-win11-demo
flatten: true
flatten_format: qcow2
to_output: win11.qcow2
out_format: qcow2
guest_os: windows
virtio_drivers_dir: /tmp/virtio-win
win_driver_iso: ~/Downloads/virtio-win-0.1.285.iso
emit_domain_xml: true
vm_name: win11-demo
memory: 4096
vcpus: 2
machine: q35
uefi: true
win_stage: bootstrap          # SATA disk (safe first boot)
win_hyperv: true              # Hyper-V enlightenments
win_localtime_clock: true
libvirt_test: true           # Boot VM after conversion
timeout: 90
```

```
keep_domain: true
```

Single command: `sudo ./h2kvmctl --config win11-demo.yaml` — handles everything: flatten split VMDKs, offline registry fixes, convert, generate Windows domain XML, boot test.

Step 3: Migration Pipeline

STEP 3a — Sanity Checks (passed)

```
18:57:31 Sanity check 1/5: args...
18:57:31 Sanity check 2/5: tools...
18:57:31 Sanity check 3/5: disk space...
18:57:31 Sanity check 4/5: permissions...
18:57:31 Sanity check 5/5: network...
18:57:31 ✓ Sanity checks passed
```

STEP 3b — Flatten Split VMDK to QCOW2 (~70s)

```
18:57:31 Flatten snapshot chain
18:57:31 Flattening via qemu-img convert (single self-contained image)
18:57:31 Flattening progress: 0.3% ... 25% ... 50% ... 75% ... 85.4%
18:58:41 ✓ Flattened: working-flattened.qcow2 (15 GiB virtual, 13.1 GiB actual)
```

STEP 3c — Offline Guest Fixes (VMCraft Engine)

```
18:58:42 Opening offline image via qemu-nbd
18:58:42 Connected to /dev/nbd0
18:58:42 Detected Windows OS on /dev/nbd0p3
18:58:42   Product: Windows 11 Home
18:58:42   OS Name: Windows 11
18:58:42   Version: 10.0
18:58:42   Distribution: windows
18:58:42 ✓ Offline fixes complete
```

STEP 3d — Convert to Final QCOW2 (~40s)

```
18:58:59 Convert image → win11.qcow2
18:58:59 Converting: working-flattened.qcow2 → win11.qcow2
          (in_format=qcow2, out_format=qcow2, compress=False)
18:58:59 Conversion progress: 0.2% ... 25% ... 50% ... 75% ... 85.4%
```

18:59:39 ✓ Validated: win11.qcow2

Step 4: Windows Registry Fixes & Boot

STEP 4a — Domain XML Emission (Windows Bootstrap)

```
18:59:23 emit_domain_xml: guest=windows uefi=True name=win11-demo
18:59:23 ✓ Domain XML: out-win11-demo/libvirt/win11-demo.xml
```

Key Windows bootstrap features:

```
Disk bus:      SATA (sda)    - Safe first boot without VirtIO storage driver
Driver CDROM:  virtio-win-0.1.285.iso (sdc)
Hyper-V:       relaxed + vpic + spinlocks
Clock:         localtime + hypervclock timer
Guest agent:   VirtIO channel (org.qemu.guest_agent.0)
Video:         QXL
Graphics:      SPICE
```

STEP 4b — Libvirt Smoke Test (SATA Bootstrap)

```
18:59:23 Libvirt smoke test (emitted XML)
18:59:23 Domain: win11-demo
18:59:23 XML: out-win11-demo/libvirt/win11-demo.xml
18:59:24 Domain reached RUNNING state after 0s.
✓ Smoke test passed: domain is RUNNING
```

STEP 4c — VirtIO Deployment (offline registry)

Post-boot VirtIO deployment (runs after smoke test confirms SATA boot):

- ✓ storahci enabled as boot-critical (Start=0) - AHCI/SATA driver for KVM
- ✓ 1 VMware service disabled (vmtoolsd)
- ✓ RunOnce batch staged for VirtIO driver install on next boot

STEP 5 — Windows 11 OOB on KVM

virsh dominfo win11-demo

```
Name:      win11-demo
State:     running
```

Windows 11 OOB — Country Selection

 Windows 11 OOB on KVM

CPU(s): 2
Max memory: 4194304 KiB
Machine: pc-q35-10.1
Firmware: UEFI (OVMF)
Disk bus: SATA (sda)
CDROM: virtio-win-0.1.285.iso (sdc)
Network: VirtIO (virbr0)
Hyper-V: enabled

Windows 11 Home — OOBE country selection
Zero BSOD — migration successful

Step 6: Output & Summary

STEP 6 — Generated Output Files

out-win11-demo/		
win11.qcow2	14 GB	Final converted image (SATA, UEFI)
win11-demo.VARS.fd	128 KB	UEFI NVRAM variables
libvirt/		
win11-demo.xml	1.7 KB	Libvirt domain XML (Q35, SATA, OVMF, Hyper-V)
work/		
working-flattened.qcow2	13 GB	Intermediate flattened image

Key Domain XML Features (Windows Bootstrap)

```
<domain type='kvm'>
  <name>win11-demo</name>
  <description>Windows UEFI domain (bootstrap): SATA bootstrap</description>
  <memory unit='MiB'>4096</memory>
  <vcpu>2</vcpu>
  <os>
    <type arch='x86_64' machine='q35'>hvm</type>
    <loader type='pflash'>OVMF_CODE.fd</loader>
    <nvram template='OVMF_VARS.fd' />
  </os>
  <features><acpi/><apic/>
    <hyperv><relaxed/><vapic/><spinlocks retries='8191' /></hyperv>
  </features>
  <clock offset='localtime'><timer name='hypervclock' /></clock>
  <devices>
    <disk bus='sata' dev='sda' />          <!-- SATA bootstrap -->
    <disk device='cdrom'>virtio-win.iso</disk> <!-- Driver CDROM -->
    <interface type='network' model='virtio' />
    <channel type='virtio' name='org.qemu.guest_agent.0' />
  </devices>
</domain>
```

Migration Pipeline Summary

#	Step	Action	Duration	Result
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1	Sanity checks	Tools, disk space, permissions	<1s	5/5 passed
2	Flatten	Split VMDK (4 extents) → QCOW2	~70s	working-flattened.qcow2 (15 GiB)
3	Offline fixes	OS detect, VMCraft engine	~4s	Windows 11 Home detected
4	Convert	Working → final QCOW2	~40s	win11.qcow2 (14 GB)
5	Emit XML	Windows bootstrap domain XML	<1s	Q35, SATA, OVMF, Hyper-V, virtio-win CDROM
6	Boot test	virsh define + start (emitted XML)	<1s	RUNNING state confirmed
7	Registry fixes	storahci=0, VMware disabled, RunOnce	~18s	SATA boot-critical, VirtIO staged
8	OOBE	Windows 11 first-run setup	~60s	Country selection reached, zero BSOD

Migration Complete — Zero BSOD, Zero Manual Intervention

VMware Windows 11 Home split VMDK (13.2 GB, 4 extents) → KVM QCOW2 with SATA + UEFI + Q35

Offline fixes: storahci boot-critical, VMware services disabled, VirtIO driver RunOnce staged

VM booted on KVM via SATA bootstrap, reached Windows 11 OOBE (country selection)

Total pipeline: ~2 minutes. Single command. Zero BSOD. Zero manual steps.